CHEST

The Effect of Respiratory Motion on Pulmonary Nodule Location During Electromagnetic Navigation Bronchoscopy

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BACKGROUND: Electromagnetic navigation has improved the diagnostic yield of peripheral bronchoscopy for pulmonary nodules. For these procedures, a thin-slice chest CT scan is performed prior to bronchoscopy at full inspiration and is used to create virtual airway reconstructions that are used as a map during bronchoscopy. Movement of the lung occurs with respiratory variation during bronchoscopy, and the location of pulmonary nodules during procedures may differ significantly from their location on the initial planning full-inspiratory chest CT scan. This study was performed to quantify pulmonary nodule movement from full inspiration to end-exhalation during tidal volume breathing in patients undergoing electromagnetic navigation procedures.

METHODS: A retrospective review of electromagnetic navigation procedures was performed for which two preprocedure CT scans were performed prior to bronchoscopy. One CT scan was performed at full inspiration, and a second CT scan was performed at end-exhalation during tidal volume breathing. Pulmonary lesions were identified on both CT scans, and distances between positions were recorded.

RESULTS: Eighty-five pulmonary lesions were identified in 46 patients. Average motion of all pulmonary lesions was 17.6 mm. Pulmonary lesions located in the lower lobes moved significantly more than upper lobe nodules. Size and distance from the pleura did not significantly impact movement.

CONCLUSIONS: Significant movement of pulmonary lesions occurs between full inspiration and end-exhalation during tidal volume breathing. This movement from full inspiration on planning chest CT scan to tidal volume breathing during bronchoscopy may significantly affect the diagnostic yield of electromagnetic navigation bronchoscopy procedures.

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ABBREVIATIONS: 3-D = three-dimensional; ENB = electromagnetic navigation bronchoscopy; EXP = expiratory; INSP = inspiratory

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With the increasing use of chest CT scans for a myriad of chest disorders, solitary pulmonary nodules have become a common finding. In addition, the national lung cancer screening trial demonstrated a significant reduction in lung cancer-related mortality among high-risk patients screened with low-dose chest CT scans.[1] With the potential widespread adoption of lung cancer screening, physicians can expect to see patients with increasing numbers of pulmonary nodules presenting to their practices, some of whom will require diagnostic procedures.

Technological advancements, such as electromagnetic navigation bronchoscopy (ENB), have improved the diagnostic yield of bronchoscopy over conventional bronchoscopic approaches.[2,3] Several studies, including a large meta-analysis, have demonstrated diagnostic yields of 67% using ENB.[4] This offers a significant improvement over conventional bronchoscopic techniques, which have traditionally had a diagnostic yield of < 20% for smaller, peripheral lung nodules.[5,6] In spite of growing experience with navigation technology, further improvements in diagnostic yield have not been observed.

Electromagnetic navigation makes use of a reference, thin-slice, chest CT scan to create a virtual airway reconstruction. An electromagnetic sensor advanced into the airways during bronchoscopy is then paired with the airway reconstruction using registration points, external fiducial markers, or both.[7] Prior to bronchoscopy, the reference CT scan is obtained by instructing patients to take a deep breath and hold at full inspiration, where the physical state of the lung approximates total lung capacity.

One observation is that bronchoscopy is a dynamic process performed in patients who are breathing either spontaneously or in a controlled fashion under the influence of procedural sedation. Accordingly, targeted lung nodules are subject to movement due to respiratory motion during bronchoscopy, where the physical state of the lung is likely closer to tidal volume than to total lung capacity.

Pulmonary nodule movement from full inspiration to end-exhalation during tidal volume breathing is unknown and has not been described. The purpose of this study is to quantify pulmonary lesion movement within the lung between full inspiration and end-exhalation with tidal volume breathing for electromagnetic navigation bronchoscopy procedures.

Materials and Methods

This was a retrospective review of deidentified patient datasets of electromagnetic navigation cases. Individual datasets consisted of full inspiratory chest CT scans and end-expiratory chest CT scans performed during tidal volume breathing prior to electromagnetic navigation procedures. This study was evaluated by the Institutional Review Board and was considered exempt from full review, as no patient identification was associated with the datasets.

CT Scan Protocol

CT scans were performed using a slice thickness ranging from 0.5 to 1.0 mm and a scan time of 10 to 15 s. Patients were instructed to breathe normally (at tidal volume) and then take a deep breath (at full inspiration) with arms raised above their head (inspiratory CT scan). Patients were then instructed to breathe normally, and a CT scan was taken while patients performed a breath hold at the end of expiration (expiratory CT scan) during normal tidal breathing with arms to their sides. These two scans were used as the inspiratory (INSP)-expiratory (EXP) CT scan pair. The same pulmonary lesion was identified on each INSP-EXP CT scan pair, and two independent investigators (A. C., B. F.) confirmed that the pulmonary lesion identified on each INSP-EXP CT scan pair represented the same lesion.

INSP-EXP CT Scan Pairing

INSP-EXP CT scan pairs were aligned using two methods to determine the respiratory motion between the INSP state and EXP state. The INSP-EXP CT scan pairs were first aligned using the main carina as a common point of translation between datasets, and the physical three-dimensional (3-D) motion was calculated (total movement). Motion in the X direction equated to medial and lateral movement, motion in the Y direction equated to anterior and posterior movement, and motion in the Z direction equated to cranial and caudal movement within each patient. Anterior and posterior points were defined based on their location relative to the main carina. Lumen registration was also implemented between the INSP and EXP scan using the airway trees in each of the scans to compensate for the shape change of the lung and associated airways. Airways were segmented from the INSP scan to provide a robust airway tree, and nonrigid deformable registration was applied to the dataset to define the segmented airway tree in the EXP scan using the SPiN Planning 2.0 workstation (Veran Medical Technologies). Lumen registration was then applied to align the INSP and EXP datasets and calculate the nonlinear 3-D motion of the lung.

Measurements

Pulmonary lesion size was recorded as the largest diameter on axial CT imaging, and movement was measured from the lesion center during full inspiration to the nodule center at end-exhalation during tidal volume breathing. Respiratory movement in the X, Y, and Z directions was calculated as a vector where movement $(m) = \sqrt{(x^2 + y^2 + z^2)}$; this was reported as the physical 3-D movement of the lung lesion. Lesion movement reported is based on the physical motion of lesions between full inspiration and end-exhalation during tidal volume breathing when the inspiratory scan is overlaid upon the end-exhalation scan using the main carina as the common point of translation.

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Results

Baseline Characteristics

Eighty-five pulmonary lesions were identified in 46 patients, providing 46 INSP-EXP datasets. The size of pulmonary lesions ranged from 6 to 42 mm, with a mean diameter of 16.6 mm using an axial image to measure lesions in their longest axis. Distribution of pulmonary lesions was as follows: 23 in the right upper lobe, 21 in the left upper lobe, 21 in the right lower lobe, and 20 in the left lower lobe. Distance from the pleura was used to identify relative location of pulmonary lesions in the lung parenchyma and was measured as the shortest measured distance to the pleura anteriorly, laterally, or posteriorly (Fig 1). Seventeen lesions were adherent to the pleura, 22 lesions were within 10 mm of the pleura, 19 lesions were 11 to 20 mm from the pleura, 12 lesions were 21 to 30 mm from the pleura, four lesions were 31 to 40 mm from the pleura, four lesions were 41 to 50 mm from the pleura, and six lesions were > 50 mm from the pleura (Table 1).

Movement by Lobe

The average motion between full inspiration and tidal volume expiration was 17.6 mm. By anatomic location, the average movement of pulmonary lesions observed in the right upper lobe was 12.2 mm, 10.6 mm in the left upper lobe, 25.3 mm in the right lower lobe, and 23.8 mm in the left lower lobe (Table 2). Movement in the Y (anterior-posterior) and Z (cranial-caudal) axis accounted for the majority of the physical 3-D movements (Table 2).

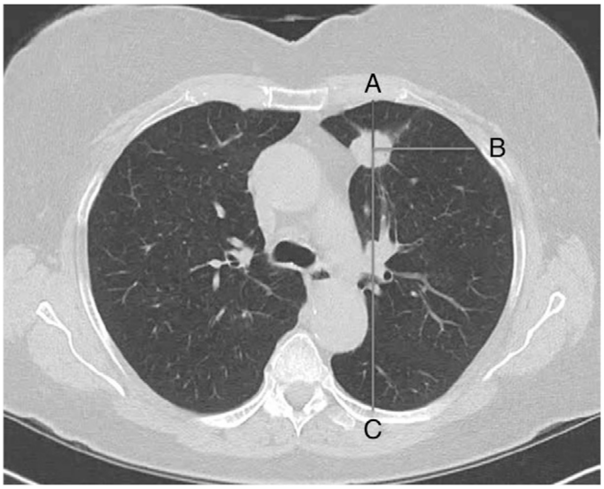

Figure 1 – *Distance from pleura measured anteriorly (A), laterally (B), or posteriorly (C). This nodule would be measured in the anterior direction as the shortest distance to A, B, or C.*

TABLE 1] Baseline Characteristics

Characteristic	No.
Patient demographics, y	
Male	24
Female	22
Age range	21-88
Mean age	68.2
Nodules per patient	
1	27
2	9
3	6
> 3	4
Nodule size, mm	
< 10	27
11-20	37
21-30	14
31-40	5
> 40	2
Nodule location	
Right upper lobe	23
Left upper lobe	21
Right lower lobe	21
Left lower lobe	20
Distance from pleura, mm	
0[a]	17
1-10	22
11-20	19
21-30	12
31-40	4
41-50	4
> 50	6

[a]Indicates that nodule was adherent to pleura.

Movement Relative to Distance From Pleura

By distance from pleura, the average movement of pulmonary lesions adherent to the pleura was 19.0 mm, for lesions 1 to 10 mm from the pleura the motion was 14.9 mm, for lesions 11 to 20 mm from the pleura the motion was 21.2 mm, for lesions 21 to 30 mm from the pleura the motion was 15.6 mm, for lesions 31 to 40 mm for the pleura the motion was 8.74 mm, for lesions 41 to 50 mm from the pleura the motion was 16.9 mm, and for lesions > 50 mm from the pleura the motion was 22.5 mm (Table 3).

Movement Relative to Lesion Size

Lesion movement recorded by size was as follows: nodules 6 to 10 mm moved 15.71 mm on average, nodules

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TABLE 2] Nodule Movement by Location

Motion, mm	Nodules, No.			
	RUL	LUL	RLL	LLL
< 6	3	6	1	1
6-8	3	2	1	0
8-10	4	4	0	1
10-12	3	3	2	0
12-14	3	2	0	2
14-16	2	2	2	1
16-18	2	0	3	1
18-20	0	0	1	3
20-22	1	1	0	0
22-24	0	0	0	1
24-26	1	1	1	3
26-28	0	...	3	0
28-30	1	...	0	0
30-32	...	...	1	1
32-34	...	...	0	3
34-36	...	...	3	0
36-38	...	...	0	1
38-40	...	...	1	1
40-60	...	...	2	1

Target motion on the y-axis and number of nodules per lobe that moved the corresponding distance, separated by lobe. LLL = left lower lobe; LUL = left upper lobe; RLL = right lower lobe; RUL = right upper lobe.

TABLE 3] Nodule Motion by Distance From Pleura

Motion, mm	Distance, mm						
	0	1-10	11-20	21-30	31-40	41-50	> 50
< 6	1	1	3	3	1	1	1
6-8	2	1	1	1	0	0	1
8-10	0	4	1	1	2	0	1
10-12	1	3	1	1	1	1	0
12-14	3	2	0	2	...	0	0
14-16	2	2	3	0	...	0	0
16-18	2	3	1	0	...	0	0
18-20	0	1	1	1	...	1	0
20-22	1	1	0	0	...	0	0
22-24	1	0	0	0	...	0	0
24-26	1	4	1	0	...	0	0
26-28	1	...	0	1	...	0	1
28-30	0	...	1	0	...	0	0
30-32	0	...	2	0	...	0	0
32-34	1	...	0	1	...	1	0
34-36	0	...	1	0	...	...	2
36-38	0	...	0	0	...	...	1
38-40	0	...	2	0	...	...	...
40-60	1	...	1	1	...	...	...

Target motion on the y-axis and distance from the pleura on the x-axis. "0 mm" distance from the pleura indicates that the target nodule was adherent to the pleura.

11 to 15 mm moved 16.85 mm, nodules 16 to 20 mm moved 17.64 mm, nodules 21 to 25 mm moved 22.06 mm, nodules 26 to 30 mm moved 24.98 mm, and lesions > 30 mm moved 15.93 mm (Table 4).

Discussion

This study demonstrates the significant variation in pulmonary lesion motion between different phases of respiration seen on CT scan. A breath hold maneuver at full inspiration in this study approximates the physical state of the lung at total lung capacity while the expiratory CT scan was performed at end-exhalation during "normal" tidal volume breathing. Peripheral lesions moved 17.6 mm on average between these two respiratory phases, and lower lobe lesions moved approximately two times the distance of upper lobe lesions (Fig 2). Lesion size and distance from the pleura did not appear to have significant effects on movement.

ENB has become a recommended diagnostic tool for the evaluation of patients with pulmonary nodules that are difficult to reach with conventional bronchoscopy.[8] This procedure relies on static chest CT scan information to create airway reconstructions used as image guidance during bronchoscopy. As an electromagnetic sensor is passed through the tracheobronchial tree during bronchoscopy, this sensor is paired to the airway reconstruction, which is used as a map.

Reference chest CT scans for electromagnetic navigation procedures are often performed at full inspiration, where smaller, peripheral airways are more visible and available for procedure planning. During bronchoscopy, patients are unlikely to be at a full inspiratory state and are more likely to be taking tidal volume breaths. Given this discrepancy between the planning chest CT scan respiratory state and the respiratory state during bronchoscopy, the relative location of a targeted lung nodule on the inspiratory planning chest CT scan may differ significantly from the location of the nodule within the lung during bronchoscopy (Fig 3).

Transformation of the lung with respiratory variation is an elastic and nonrigid process that is not predictable using linear translations.[9] That is to say that if the main carina is found to move in a cranial fashion by 15 mm between inspiration and expiration, the same movement cannot be predicted to occur for a right lower lobe

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TABLE 4] Nodule Motion by Size

	Nodule Size, mm					
Motion, mm	6-10	11-15	16-20	21-25	26-30	> 30
< 6	3	4	2	0	0	2
6-8	3	0	1	0	2	0
8-10	2	3	2	1	1	0
10-12	3	2	2	0	0	1
12-14	1	1	1	2	1	1
14-16	4	1	2	0	0	0
16-18	5	0	1	0	0	0
18-20	1	1	1	1	0	0
20-22	0	1	0	0	0	1
22-24	0	0	0	1	0	0
24-26	0	1	3	1	0	1
26-28	2	1	0	0	0	0
28-30	0	0	0	0	0	1
30-32	0	2	0	0	0	0
32-34	1	1	0	1	0	0
34-36	1	1	1	0	0	0
36-38	0	0	1	0	0	0
38-40	1	0	1	0	0	0
40-60	0	0	0	1	2	0

Target motion on the y-axis and nodule size on the x-axis.

nodule. Additionally, the presence of a bronchoscope in the lung periphery may alter these mechanics, leading to further physical displacement of targeted lesions.[10]

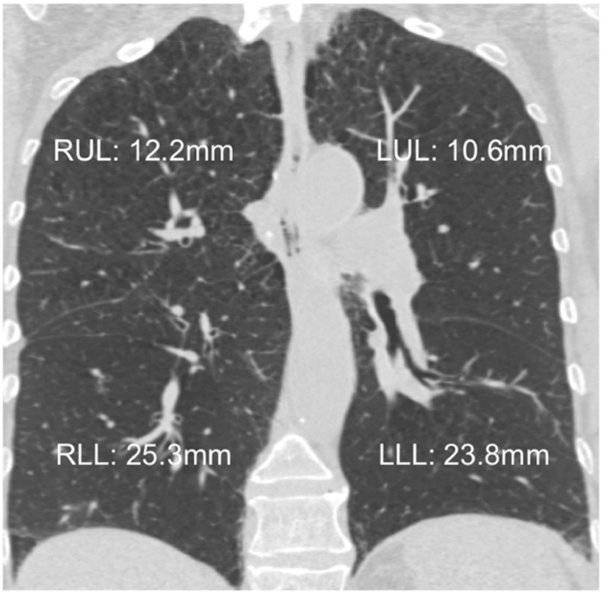

Figure 2 – *Nodule movement by lobe. Average nodule movement per lobe. LLL = left lower lobe; LUL = left upper lobe; RLL = right lower lobe; RUL = right upper lobe.*

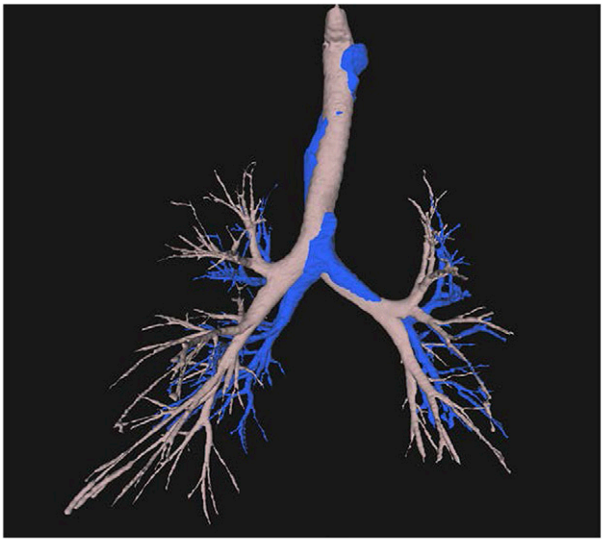

Figure 3 – *Inspiratory and expiratory chest CT scan. Tracheobronchial tree on full inspiration in gray and on end-exhalation from tidal volume in blue.*

For successful navigation to occur with ENB, it is intuitive that the physical shape of the lung and tracheobronchial tree during bronchoscopy be as similar as possible to the shape of the lung and tracheobronchial tree captured on the planning chest CT scan. If the two images are discordant, then navigational error may be introduced into the system. The planning CT scan is used to define the bronchoscopist's route to the targeted lung lesion; if the peripheral airways on the chest CT scan are not representative of real-time peripheral airways during bronchoscopy, then this route may not be accurate.

Characterization of pulmonary nodule movement with respiratory variation during tidal volume breathing has been described in radiation oncology, where high doses of radiation are delivered to targeted areas of the lung in spontaneously breathing patients. During tidal volume breathing, nodule movement on average of 10 mm has been described, with lower lobe lesions demonstrating more movement than upper lobe lesions.[11-13] Real-time procedural imaging such as cone beam CT scan may be used to track tumor motion during stereotactic body radiation therapy and guide therapy, thereby minimizing radiation to adjacent, healthy lung. The findings in this study demonstrate that significantly more nodule movement occurs between full inspiration and end-exhalation with tidal volume breathing. Unlike image guidance used during stereotactic body radiation therapy to track nodule movement, no mechanism exists during electromagnetic navigation procedures to track movement of pulmonary nodules from the full

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inspiratory state on the planning chest CT scan to their position while breathing during bronchoscopy. Accordingly, there is no compensatory mechanism available for bronchoscopists to account for nodule movement during navigational procedures. In light of this, the use of additional technology, such as radial probe endobronchial ultrasound, to confirm localization of the target lesion prior to biopsy may be advantageous.

ENB procedures have been likened to global positioning systems to assist travelers with locating their destination. One important difference between these two systems is that global positioning system devices receive continuous positioning feedback regarding their location on a map that is continuously updated; ENB systems do not update the "map" during bronchoscopy, and, therefore, the electromagnetic sensor is paired with the planning map that has been generated from the original, full inspiratory chest CT scan. As there is no continuous updating of the "map" during bronchoscopy, the initial planning chest CT scan should reflect the physical state of the lung during bronchoscopy as closely as possible.

This study is the first that we are aware of that characterizes lung nodule movement from full inspiration to end-exhalation during normal tidal volume breathing. A limitation of this study is that it only estimates the state of the lung during bronchoscopy. Accordingly, precise comparisons between the state of the lung at full inspiration and during bronchoscopy cannot be made.

Movement of the lung with respiratory variation during procedures is an observation routinely made during bronchoscopy. The magnitude of this movement between different phases of respiration has not been clinically relevant previously, as bronchoscopic procedures had not incorporated reference images from CT scans or other radiographic images for guidance purposes. With the development of advanced image-guided diagnostic bronchoscopy, including virtual bronchoscopy and electromagnetic navigation, this information becomes clinically relevant and may have a significant impact on diagnostic outcome.

In summary, this study demonstrates the significant movement of peripheral pulmonary lesions that occurs between full inspiration at the time of planning chest CT scan to end-exhalation during tidal volume breathing. Clinically, this finding may account for some of the challenges seen with improving diagnostic yields for electromagnetic navigation procedures. The location of pulmonary lesions on full inspiratory planning chest CT scan does not reflect the actual position of these lesions at the time of bronchoscopy. Future endeavors to improve the diagnostic yield of these procedures may need to account for this.

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Author contributions: A. C. takes responsibility for (is the guarantor of) the content of the manuscript, including the data and analysis. A. C., N. P., B. F., and G. A. S. contributed to study conception and design, or acquisition of data, or analysis and interpretation of data; drafted the submitted article or revised it critically for important intellectual content; provided final approval of the version to be published; and agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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References

1. Aberle DR, Adams AM, Berg CD, et al; National Lung Screening Trial Research Team. Reduced lung-cancer mortality with low-dose computed tomographic screening. *N Engl J Med*. 2011;365(5):395-409.
2. Gildea TR, Mazzone PJ, Karnak D, Meziane M, Mehta AC. Electromagnetic navigation diagnostic bronchoscopy: a prospective study. *Am J Respir Crit Care Med*. 2006;174(9):982-989.
3. Eberhardt R, Anantham D, Herth F, Feller-Kopman D, Ernst A. Electromagnetic navigation diagnostic bronchoscopy in peripheral lung lesions. *Chest*. 2007; 131(6):1800-1805.
4. Wang Memoli JS, Nietert PJ, Silvestri GA. Meta-analysis of guided bronchoscopy for the evaluation of the pulmonary nodule. *Chest*. 2012;142(2): 385-393.
5. Baaklini WA, Reinoso MA, Gorin AB, Sharafkaneh A, Manian P. Diagnostic yield of fiberoptic bronchoscopy in evaluating solitary pulmonary nodules. *Chest*. 2000;117(4):1049-1054.
6. van 't Westeinde SC, Horeweg N, Vernhout RM, et al. The role of conventional bronchoscopy in the workup of suspicious CT scan screen-detected pulmonary nodules. *Chest*. 2012;142(2):377-384.
7. Turcza P, Duplaga M. Navigation systems based on registration of endoscopic and CT-derived virtual images for bronchofiberoscopic procedures. *Stud Health Technol Inform*. 2004;105: 253-263.
8. Rivera MP, Mehta AC, Wahidi MM. Establishing the diagnosis of lung cancer: diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2013;143(5_suppl):142S-165S.
9. Shirzadi Z, Sadeghi-Naini A, Samani A. Toward in vivo lung's tissue incompressibility characterization for tumor motion modeling in radiation therapy. *Med Phys*. 2013;40(5):051902.
10. Seppenwoolde Y, Shirato H, Kitamura K, et al. Precise and real-time measurement of 3D tumor motion in lung due to breathing and heartbeat, measured during radiotherapy. *Int J Radiat Oncol Biol Phys*. 2002;53(4):822-834.
11. Keall P, Vedam S, George R, et al. The clinical implementation of respiratory-gated intensity-modulated radiotherapy. *Med Dosim*. 2006;31(2): 152-162.
12. Shen G, Wang YJ, Sheng HG, et al. Double CT imaging can measure the respiratory movement of small pulmonary tumors during stereotactic ablative radiotherapy. *J Thorac Dis*. 2012;4(2): 131-140.
13. Leira HO, Langø T, Sorger H, Hofstad EF, Amundsen T. Bronchoscope-induced displacement of lung targets: first in vivo demonstration of effect from wedging maneuver in navigated bronchoscopy. *J Bronchology Interv Pulmonol*. 2013; 20(3):206-212.

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